

ESWAR

Aspiring DevOps & Cloud Engineer | B.Tech CSE | AWS | Docker | CI/CD | Linux

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TARGET OPPORTUNITIES & AVAILABILITY

Actively Seeking: Research-Oriented Work, Internships & Paid Opportunities

- Focus: Paid Engineering Internships (DevOps/Cloud/Full Stack) & Research-Oriented Projects
- Research Focus: Digital Twins, Predictive ML Models (0.83 ROC-AUC), Geospatial Big Data (23K+ assets)

PROFESSIONAL SUMMARY

Motivated B.Tech Computer Science Engineering student with hands-on experience in DevOps, Cloud Computing, AWS, Docker, CI/CD pipelines, Linux, Git/GitHub, FastAPI, React, PostgreSQL/PostGIS, and Machine Learning integration. Experienced in developing end-to-end research-grade projects involving cloud deployment, infrastructure monitoring, digital twins, geospatial systems, and AI-based risk prediction. Actively seeking research-oriented work, internships, and paid internship opportunities to apply automation, deployment, infrastructure, and software engineering skills.

EDUCATION

Bachelor of Technology - Computer Science and Engineering

NRI Institute of Technology | Andhra Pradesh, India

Expected: 2028

CGPA: 8.97 / 10

TECHNICAL SKILLS

Cloud & DevOps:	AWS (EC2, S3, CodePipeline, CodeBuild, CodeDeploy), Docker, Docker Compose, CI/CD, Nginx
Operating Systems:	Linux, Ubuntu, Windows, PowerShell, Bash
Programming & Backend:	Python, Java, SQL, JavaScript, TypeScript, FastAPI, REST APIs
Frontend & Database:	React, Vite, Tailwind CSS, PostgreSQL, PostGIS
GIS, Digital Twin & ML:	QGIS, Leaflet, CesiumJS, Three.js, OpenStreetMap, Scikit-learn, Pandas, NumPy, ETL
Developer Tools:	VS Code, Git, GitHub, Docker Desktop, Postman

FEATURED PROJECTS

SIMRAS - Smart Infrastructure Monitoring and Risk Assistance System [R&D Project]

[AI](#) | [Digital Twin](#) | [GIS](#) | [FastAPI](#) | [React](#) | [PostGIS](#) | [ML Pipelines](#) | [GitHub: github.com/eswarc440-lgtm/SIMRAS](#)

- Engineered an end-to-end infrastructure intelligence platform monitoring dams, bridges, and transport corridors.
- Built a GIS-based infrastructure registry with PostGIS and OpenStreetMap; analyzed 23,000+ bridge records.
- Developed predictive ML pipelines achieving held-out ROC-AUC of ~0.83 for bridge structural risk forecasting.
- Designed interactive 3D Digital Twin visualization utilizing Three.js and CesiumJS geospatial tiles.
- Implemented strict evidence-policy gates cleanly separating authoritative ground truth from model inferences.
- Deployed multi-container architecture using Docker, FastAPI, React, and PostgreSQL/PostGIS.

AWS Automated CI/CD Web Application Deployment

[AWS](#) | [CodePipeline](#) | [CodeBuild](#) | [CodeDeploy](#) | [EC2](#) | [S3](#) | [Nginx](#) | [GitHub: github.com/eswarc440-lgtm/myaws](#)

- Designed and deployed an automated CI/CD pipeline triggering automated build and deploy on GitHub commits.
- Provisioned Ubuntu EC2 instance, CodeDeploy Agent, and Nginx web server reverse-proxy configuration.
- Utilized Amazon S3 for deployment artifact staging and versioned build storage.
- Automated release lifecycle reducing manual deployment intervention to zero with production log monitoring.

Smart Disaster Resource Allocation System

[Algorithmic Decision Support](#) | [Python](#) | [PostgreSQL](#) | [Resource Scheduling](#)

- Designed algorithmic allocation models to distribute relief resources dynamically during emergencies.
- Formulated priority queuing and constraint-satisfaction algorithms based on location urgency and shelter demand.

CERTIFICATIONS & PROFESSIONAL COURSES

NPTEL - Cloud Computing

- Cloud architecture, virtualization, distributed systems, resource orchestration, and service models.

NPTEL - Internet of Things (IoT)

- Sensor networks, connectivity protocols, embedded intelligence, and telemetry processing.

KEY STRENGTHS & CORE COMPETENCIES

- Strong problem-solving and automated pipeline debugging ability in Linux and cloud environments.
- Practical hands-on experience taking systems from code commit to automated cloud release.
- Experience processing large real-world geospatial datasets (23,000+ asset records).
- Full-stack ability connecting frontend (React), backend (FastAPI), database (PostGIS), and ML models.